MSG No. 83.49 P_C4/m [Type IV, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `4/m..`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,4,5,6,7,8]`
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[9,10,11,12,13,14,15,16]`

Table 2: Wyckoff site: `2b`, site symmetry: `4/m..`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[9,10,11,12,13,14,15,16]`

Table 3: Wyckoff site: `2c`, site symmetry: `4'/m..`

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	`[1,4,5,8,10,11,14,15]`
2	$\left[\frac{1}{2}, 0, 0\right]$	`[2,3,6,7,9,12,13,16]`

Table 4: Wyckoff site: `2d`, site symmetry: `4'/m..`

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,4,5,8,10,11,14,15]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[2,3,6,7,9,12,13,16]`

Table 5: Wyckoff site: `4e`, site symmetry: `2'/m..`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[1,8,12,13]`
2	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[2,7,11,14]`
3	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[3,6,10,15]`
4	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[4,5,9,16]`

Table 6: Wyckoff site: `4f`, site symmetry: `2'/m..`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[1,8,12,13]`
2	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[2,7,11,14]`
3	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[3,6,10,15]`
4	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[4,5,9,16]`

Table 7: Wyckoff site: `4g`, site symmetry: `4..`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,3,4]`
2	$[0,\ 0,\ -z]$	`[5,6,7,8]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[9,10,11,12]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[13,14,15,16]`

Table 8: Wyckoff site: `4h`, site symmetry: `4'..`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,4,10,11]`
2	$\left[\frac{1}{2},\ 0,\ z\right]$	`[2,3,9,12]`
3	$\left[0,\ \frac{1}{2},\ -z\right]$	`[5,8,14,15]`
4	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[6,7,13,16]`

Table 9: Wyckoff site: `8i`, site symmetry: `2'..`

No.	position	mapping
1	$\left[\frac{3}{4},\ \frac{1}{4},\ z\right]$	`[1,12]`
2	$\left[\frac{3}{4},\ \frac{3}{4},\ z\right]$	`[2,11]`
3	$\left[\frac{1}{4},\ \frac{1}{4},\ z\right]$	`[3,10]`
4	$\left[\frac{1}{4},\ \frac{3}{4},\ z\right]$	`[4,9]`
5	$\left[\frac{1}{4},\ \frac{3}{4},\ -z\right]$	`[5,16]`
6	$\left[\frac{1}{4},\ \frac{1}{4},\ -z\right]$	`[6,15]`
7	$\left[\frac{3}{4},\ \frac{3}{4},\ -z\right]$	`[7,14]`
8	$\left[\frac{3}{4},\ \frac{1}{4},\ -z\right]$	`[8,13]`

Table 10: Wyckoff site: `8j`, site symmetry: `m..`

No.	position	mapping
1	$[x,\ y,\ 0]$	`[1,8]`
2	$[-y,\ x,\ 0]$	`[2,7]`
3	$[y,\ -x,\ 0]$	`[3,6]`
4	$[-x,\ -y,\ 0]$	`[4,5]`
5	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ 0\right]$	`[9,16]`
6	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ 0\right]$	`[10,15]`
7	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ 0\right]$	`[11,14]`
8	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ 0\right]$	`[12,13]`

Table 11: Wyckoff site: `8k`, site symmetry: `m..`

No.	position	mapping
1	$\left[x,\ y,\ \frac{1}{2}\right]$	`[1,8]`
2	$\left[-y,\ x,\ \frac{1}{2}\right]$	`[2,7]`

continued ...

Table 11

No.	position	mapping
3	$\left[y,\, -x,\, \frac{1}{2}\right]$	[3,6]
4	$\left[-x,\, -y,\, \frac{1}{2}\right]$	[4,5]
5	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, \frac{1}{2}\right]$	[9,16]
6	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, \frac{1}{2}\right]$	[10,15]
7	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	[11,14]
8	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, \frac{1}{2}\right]$	[12,13]

Table 12: Wyckoff site: 16l, site symmetry: 1

No.	position	mapping
1	$[x,\, y,\, z]$	[1]
2	$[-y,\, x,\, z]$	[2]
3	$[y,\, -x,\, z]$	[3]
4	$[-x,\, -y,\, z]$	[4]
5	$[-x,\, -y,\, -z]$	[5]
6	$[y,\, -x,\, -z]$	[6]
7	$[-y,\, x,\, -z]$	[7]
8	$[x,\, y,\, -z]$	[8]
9	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z\right]$	[9]
10	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, z\right]$	[10]
11	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, z\right]$	[11]
12	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z\right]$	[12]
13	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, -z\right]$	[13]
14	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, -z\right]$	[14]
15	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, -z\right]$	[15]
16	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, -z\right]$	[16]